

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 11

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

#### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive <b>(1)</b></li> <li>• Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected <b>(1)</b></li> <li>• Damage to other cell parts can stop the cell functioning or kill it; low damage may be repaired by the intact nucleus <b>(1)</b></li> </ul> | 3     |
| 01.2     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Acute: effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) <b>(1)</b></li> <li>• Chronic: effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) <b>(1)</b></li> </ul>                                                                                                                                                                  | 2     |
| 01.3     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Exposure: absorption of ionising radiation from a source the person is close enough to <b>(1)</b></li> <li>• Contamination: physically carrying radioactive material on or in the body, so exposure continues wherever the person goes; sealed sources can expose but not contaminate <b>(1)</b></li> </ul>                                                                                                                                                                                  | 2     |
| 01.4     | <p><i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i></p> <ul style="list-style-type: none"> <li>• Sieverts = grays x radiation weighting factor (20 for alpha) <b>(1)</b></li> <li>• <math>0.002 \times 20 = 0.04 \text{ Sv}</math> (40 mSv) <b>(1)</b></li> </ul>                                                                                                                                                                                                     | 2     |
| 01.5     | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• The activity of a source (number of decays per second) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | 1 mark for the correct answer.<br>• $(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%) (1)                                                                                                                                                                                                                                                                                                                                                                | 1     |
| 02.2     | 1 mark for each point.<br>• A fixed fee gives no incentive to save water because the bill is the same however much is used (1)<br>• Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) (1)                                                                                                                                                                                        | 2     |
| 02.3     | Any 3 from the points below - 1 mark each, maximum 3.<br>• Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers (1)<br>• Greywater re-use: bath / shower water used to flush toilets or water gardens (1)<br>• Rainwater collection / water butts for garden use (1)<br>• Xeriscaping: planting drought-adapted species that need no watering (1)<br>• Repairing leaks / dripping taps (1) | 3     |
| 02.4     | 1 mark for each point.<br>• Old infrastructure / pipes in poor condition leak (1)<br>• Repair and good maintenance / pipe replacement / leak detection reduce wastage (1)                                                                                                                                                                                                                                                                                   | 2     |
| 02.5     | 1 mark for each point.<br>• Water is delivered directly to the plant roots / soil around each plant (1)<br>• Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff (1)                                                                                                                                                                                                                                             | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.                                                                                                                                                                                                                                                                                                                                     |       |
| 03.1     | Correct answer only.<br>• Organic (methyl) mercury (1)                                                                                                                                                                                                                                                                                                                                                                                                      | 1     |
| 03.2     | 1 mark for each point.<br>• Anaerobic microbes converted the inorganic mercury into organic methyl mercury (1)<br>• Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people (1)                                                                                                                                                                                               | 2     |
| 03.3     | 1 mark for each point.<br>• Most heavy metals are less soluble at high pH (1)<br>• So they remain immobile and are not leached into water or absorbed by organisms (1)                                                                                                                                                                                                                                                                                      | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.                                                                                                                                                                                                                                                                                                                                  |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier <b>(1)</b></li> <li>• Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground <b>(1)</b></li> <li>• Take repeated readings at each distance and calculate means (and standard deviation) to assess variability <b>(1)</b></li> <li>• Use a measure such as LA10 or Leq over a set period rather than a single peak reading <b>(1)</b></li> <li>• Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance <b>(1)</b></li> </ul> | 5     |
| <b>04.2</b> | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods <b>(1)</b></li> <li>• LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2     |
| <b>04.4</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>• Double / triple glazing <b>(1)</b></li> <li>• Acoustic insulation in walls / roof <b>(1)</b></li> <li>• Bedrooms on the side facing away from the track <b>(1)</b></li> <li>• Dense hedges / trees and earth mounds in gardens <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>05.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) <b>(1)</b></li> <li>• Magnetometry <b>(1)</b></li> <li>• Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata <b>(1)</b></li> <li>• Resistivity <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                               | 4     |
| <b>05.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Trial drilling is the most expensive technique per site <b>(1)</b></li> <li>• Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites <b>(1)</b></li> <li>• Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                      | 3     |
| <b>05.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Different minerals emit / reflect infrared at different wavelengths <b>(1)</b></li> <li>• Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>05.4</b> | <p>1 mark.</p> <ul style="list-style-type: none"> <li>Results are available immediately in the field, saving time waiting for laboratory analysis <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                           | 1     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 137; AQA spec 3.2.3.4.</p>                                                                                                                                                                                                                                                                                                                                                                    |       |
| <b>06.1</b> | <p>1 mark for the correct answer.</p> <ul style="list-style-type: none"> <li><math>7 \times 10^{11}</math> tonnes <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                           | 1     |
| <b>06.2</b> | <p>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</p> <ul style="list-style-type: none"> <li>10% of 10 000 = 1000 billion tonnes <b>(1)</b></li> <li><math>1000 / 700 \times 100 = 143\%</math> (accept 142.9) <b>(1)</b></li> </ul>                                                                                                                                                              | 2     |
| <b>06.3</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) <b>(1)</b></li> <li>Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion <b>(1)</b></li> </ul>                                                                                                                       | 2     |
| <b>06.4</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>CO<sub>2</sub> dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis <b>(1)</b></li> <li>Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons <b>(1)</b></li> <li>When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods <b>(1)</b></li> </ul> | 3     |
| <b>06.5</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Aerobic respiration uses oxygen to break down organic compounds completely to CO<sub>2</sub> (and water), releasing much more energy <b>(1)</b></li> <li>Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH<sub>4</sub>), which is later oxidised to CO<sub>2</sub> in the atmosphere <b>(1)</b></li> </ul>                                                       | 2     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.</p>                                                                                                                                                                                                                                                                                                                                                                |       |
| <b>07.1</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines <b>(1)</b></li> <li>Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost <b>(1)</b></li> </ul>                                                                                 | 2     |
| <b>07.2</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>As volume increases the surface area : volume ratio decreases <b>(1)</b></li> <li>Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage) <b>(1)</b></li> </ul>                                                                                                                                                | 2     |
| <b>07.3</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation <b>(1)</b></li> <li>Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines <b>(1)</b></li> </ul>                                                                                                                                     | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>07.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Materials with high specific heat capacity (concrete, water) warm and cool slowly <b>(1)</b></li> <li>They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating <b>(1)</b></li> </ul>                                                                                                                                                                                                                          | 2     |
| <b>07.5</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation <b>(1)</b></li> <li>Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage <b>(1)</b></li> <li>Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries <b>(1)</b></li> </ul>                                                                     | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>08.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable <b>(1)</b></li> <li>Wild species have different ranges of tolerance so cross results cannot be applied directly <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                        | 2     |
| <b>08.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Preliminary study with widely spaced pH values to find where germination occurs <b>(1)</b></li> <li>Final range from below the lowest pH giving germination to above the highest, with closer intervals <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                    | 2     |
| <b>08.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Temperature (constant, natural germination temperature) <b>(1)</b></li> <li>Light, water volume, number of seeds per dish, seed source <b>(1)</b></li> <li>Growth medium (filter paper; soil would alter pH); nutrients not needed for germination <b>(1)</b></li> <li>Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure <b>(1)</b></li> </ul>                                                                   | 4     |
| <b>08.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Surface water affects fresh mass and drying damages the plants; dry mass requires killing the plants so the same plants cannot be re-measured <b>(1)</b></li> <li>Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time <b>(1)</b></li> </ul>                                                                                                                                                                                              | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| <b>09.1</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff <b>(2)</b></li> <li>Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover <b>(2)</b></li> </ul> | 4     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 09.2     | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Tree roots that held the soil together decompose so the soil is bound less strongly (1)</li> <li>• Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
| 09.3     | <p>Any 4 from the points below - 1 mark each, maximum 4.</p> <ul style="list-style-type: none"> <li>• Sedimentation in rivers reduces flow capacity so flooding downstream is more likely (1)</li> <li>• Sedimentation in reservoirs reduces their storage volume (1)</li> <li>• Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants (1)</li> <li>• Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas (1)</li> <li>• Desertification as vegetation cannot re-establish; reduced rainfall (1)</li> <li>• Nutrients (phosphates) carried on soil particles causing eutrophication (1)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |
| 10.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine</li> <li>• High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier</li> <li>• Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline</li> <li>• Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar</li> <li>• Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes</li> <li>• Synergism with some fungicides increases toxicity to bees</li> <li>• Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind</li> <li>• Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift</li> <li>• Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control)</li> <li>• Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields</li> </ul> | 9     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10.2     | <p><i>Any 1 from the points below - 1 mark each, maximum 1.</i></p> <ul style="list-style-type: none"> <li>• Biological control (introduced predators / pathogens) <b>(1)</b></li> <li>• Predator habitats (hedgerows, beetle banks) <b>(1)</b></li> <li>• Crop rotation <b>(1)</b></li> <li>• Companion / barrier crops <b>(1)</b></li> <li>• Sterile male technique / pheromone traps <b>(1)</b></li> <li>• Genetic resistance <b>(1)</b></li> </ul> | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.</p>                                                                                                                                                                                                                                                                                                                   |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
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| 11.1     | <p><b>Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry)</li> <li>• Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs</li> <li>• Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta</li> <li>• Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination</li> <li>• Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips)</li> <li>• Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management)</li> <li>• Evaluation of which strategies are most sustainable in different regions; conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 114-128; AQA spec 3.2.2.</p> | 25    |



| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
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| 11.2     | <p><b>Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments</li> <li>• Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques</li> <li>• Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers</li> <li>• Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules</li> <li>• Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use</li> <li>• Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts</li> <li>• Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.</p> | 25    |